

MADHU RITHIKA R K

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AI Engineer and Data Scientist building real-world systems in clinical AI, inclusive fintech, and safety — with national hackathon recognition across Smartathon 2.0, SIH 2025, IIT Palakkad, and AI by HER Global Impact Challenge - India AI Impact Summit.

SKILLS

Artificial Intelligence Deep Learning, Generative AI, Large Language Models, Multi-Agent Systems, Digital Twin Modeling, Predictive Modeling, Model Optimization	Programming Languages Python, Java, R Data & Infrastructure SQL, PostgreSQL, MongoDB, Apache Kafka, Hadoop, Snowflake	Tools & Technologies Docker, Git, Tableau, Power BI, Qlik Sense Hardware & IoT ESP32, TinyML, Sensors, MQTT
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PROJECTS

- Multi Agent Clinical Workflow Orchestrator**
 - Built a secure, privacy-first EHR platform with patient-controlled access and facial authentication.
 - Developed offline AI clinical scribing and real-time decision support to automate documentation and detect drug–allergy risks.
- SalesOS.ai – Agentic AI Sales Intelligence & Revenue Acceleration Platform**
 - Built an event-driven, multi-agent AI system to autonomously score leads and accelerate deal conversions.
 - Implemented explainable decision logging and real-time revenue dashboards.
- AI Deepfake & Manipulated Content Verification System**
 - Built a multi-modal AI system to detect deepfakes using rPPG, voice scams, and malicious links with confidence scoring.
 - Enabled real-time alerts, one-tap verification, and voice warnings to prevent fraud.
- AI-Enhanced EV Digital Twin & Predictive Analytics**
 - Created an AI-powered EV digital twin to predict battery health and optimize energy efficiency.
 - Built models to forecast vehicle range and power consumption from telemetry data.
- AI-based Human Stampede Early Warning System**
 - Developed a real-time AI system to analyze crowd density and movement for early stampede risk detection.
 - Implemented a live dashboard with automated emergency alerts to authorities.
- Aarambam - Voice-First AI-Based UPI for Inclusive Finance** | Shortlisted for AI by HER: Global Impact Challenge
 - Built a multilingual voice-first AI assistant enabling UPI transactions, form filling, and banking guidance, removing literacy barriers.
 - Implemented a real-time AI scam shield that detects fraud cues and delivers instant vernacular voice warnings for secure transactions.
- IoT-Enabled Solar Powered Mine Dewatering System with Remote Monitoring** | Shortlisted for SIH'25
 - Engineered solar-powered automated mine dewatering solution with remote control and monitoring.
 - Integrated predictive maintenance and digital twin to forecast failures and improve operational reliability.

PROFESSIONAL EXPERIENCE

BYTS - Python Intern	June 2024
Developed Python-based solutions for real-world problems, strengthening problem-solving and technical presentation skills.	
Optimus Technocrates - Machine Learning Intern	September 2024 - October 2024
Engineered Python-based ML models for predictive data analysis and process automation.	

EDUCATION

Karpagam College of Engineering	September 2023 - Present
Bachelor of Technology in Artificial Intelligence and Data Science	CGPA - 9.08
St. Michael MHSS, Madurai	May 2022
HSC	Percentage - 94.67%

ACHIEVEMENTS

- India AI Impact Summit'26 AI by HER: Global Impact Challenge, Shortlisted in Top 1.5% at Global AI Summit at Delhi
- HCL Guvi Impact AI Buildathon, Top 2% National Finalist
- Smartathon 2.0 L&T National Level Hackathon, Special Mention
- Infynd Industry Hackathon, Finalist
- National level Hackathon, IIT Palakkad, Finalist
- Smart India Hackathon 2025, Shortlisted

LEADERSHIP & INVOLVEMENT

Event Organizer	03/2025 Dhruva Karpagam College of Engineering
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CERTIFICATIONS

MongoDB Basics for Students MongoDB	Python for Data Science, Programming in Java & Responsible and Safe AI Systems NPTEL	Data Analytics & Apply AI: Analyze Customer Reviews Cisco Networking Academy	Deep Learning Fundamentals & Data Visualization with Python IBM
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